

Rajalakshmi Engineering College

Name: GULZHAR AHAMAD
Email: 240801090@rajalakshmi.edu.in
Roll no: 240801090
Phone: 9043074883
Branch: REC
Department: I ECE FA
Batch: 2028
Degree: B.E - ECE

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 1_COD_Question 1

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Janani is a tech enthusiast who loves working with polynomials. She wants to create a program that can add polynomial coefficients and provide the sum of their coefficients.

The polynomials will be represented as a linked list, where each node of the linked list contains a coefficient and an exponent. The polynomial is represented in the standard form with descending order of exponents.

Input Format

The first line of input consists of an integer n , representing the number of terms in the first polynomial.

The following n lines of input consist of two integers each: the coefficient and the exponent of the term in the first polynomial.

The next line of input consists of an integer m, representing the number of terms in the second polynomial.

The following m lines of input consist of two integers each: the coefficient and the exponent of the term in the second polynomial.

Output Format

The output prints the sum of the coefficients of the polynomials.

Sample Test Case

Input: 3

2 2

3 1

4 0

3

2 2

3 1

4 0

Output: 18

Answer

```
#include<stdio.h>
```

```
#include<stdlib.h>
```

```
typedef struct poly {  
    int coeff;  
    int expon;  
    struct poly* next;  
}Node;
```

```
Node*newnode(int coeff,int expon){  
    Node* new_node=(Node*)malloc(sizeof(Node));  
    new_node->coeff=coeff;  
    new_node->expon=expon;  
    new_node->next=NULL;  
    return new_node;  
}
```

```
void insertNode(Node**head,int coeff,int expon){  
    Node*temp=*head;  
    if(temp==NULL){  
        *head= newnode(coeff,expon);
```

```

        return;
    }
    while(temp->next !=NULL){
        temp=temp->next;
    }
    temp->next=newnode(coeff,expon);
}
int main()
{
    int n,coeff,expon;
    scanf("%d",&n);

    Node* poly1;
    Node* poly2;

    for(int i=0;i<n;i++){
        scanf("%d %d",&coeff,&expon);
        insertNode(&poly2,coeff,expon);
    }
    scanf("%d",&n);

    for(int i=0;i<n;i++){
        scanf("%d %d",&coeff,&expon);
        insertNode(&poly2,coeff,expon);
    }
    int sum=0;
    while(poly1!=NULL){
        sum+=poly1->coeff;
        poly1=poly1->next;
    }
    while(poly2!= NULL){
        sum+=poly2->coeff;
        poly2=poly2->next;
    }
    printf("%d",sum);
}

```

Status : Correct

Marks : 10/10